Trees Use Hidden 'Muscles' to Stand Up Straight

From: ScienceDaily Top Environment Mined: 2026-11-01 3 min read

Trees cannot walk or stretch. Still, they can sense their own shape and change how they grow to stay upright. This sense is called **proprioception**. People once thought only animals had it. In 2012, researchers working with INRAE showed that plants have it too.

To study this hidden sense, scientists built a special test. Young trees spun slowly on a platform inside a ball of light that shone the same from every side. **Gravity** and light gave them no clues about which way to grow.

The test began with young **poplar** trees lying flat. As they grew, their **stems** curved upward. A special kind of wood, called **tension wood**, helped. It pulls like a muscle.

After about ten days, the curved trees went onto the spinning platform. Over the next weeks, their **stems** slowly became straight again. New **tension wood** had grown on the other side, pulling the opposite way, like a second muscle.

This could help trees stand straight again after storms bend them.

Word Treasure

proprioception -- Proprioception is the sense that lets a living thing feel its own shape and position.

tension wood -- Tension wood is a special kind of wood that pulls like a muscle to help a tree change how it grows.

Gravity -- Gravity is the force that pulls things down toward the Earth.

poplar -- A poplar is a kind of fast-growing tree, and young poplars were used in the test.

stems -- A stem is the main stalk of a plant or young tree that holds it up.

Background you need

Trees cannot walk or stretch, yet they can sense their own shape and change how they grow to stay upright.

A special kind of wood called tension wood pulls like a muscle, helping a tree bend its growth one way or the other.

Learning this could help trees stand straight again after storms bend them.

Break it down

WHO: Scientists and researchers working with INRAE, studying young poplar trees.

WHAT: They studied how trees sense their own shape and straighten themselves using tension wood.

WHERE: In a special test, with trees spun slowly on a platform inside a ball of light that shone the same from every side.

WHY: Gravity and light gave the trees no clues about which way to grow, so scientists could study the trees' hidden sense on its own.

Why it matters

Understanding trees helps children care for the trees in parks, gardens and forests that they play in and enjoy.

Test what you remember

1. What is the trees' sense of their own shape called?

- (A) Tension wood
- (B) Proprioception
- (C) Gravity
- (D) INRAE

2. In 2012, researchers working with which group showed that plants have this sense too?

- (A) A spinning platform
- (B) A ball of light
- (C) INRAE
- (D) Young poplar trees

3. Which trees were used at the start of the test, lying flat?

- (A) Trees in a forest
- (B) Young poplar trees
- (C) Trees with no stems
- (D) Storm-bent trees

4. What helped the young trees' stems curve upward as they grew?

- (A) Tension wood
- (B) A second platform
- (C) Gravity and light
- (D) Spinning faster

5. After about ten days, where did the curved trees go?

- (A) Onto the spinning platform
- (B) Back outside
- (C) Flat on the ground again
- (D) Into a storm

6. How could this hidden sense help trees in nature?

- (A) It removes gravity
- (B) It lets trees walk to new places
- (C) It makes light shine the same from every side
- (D) It could help them stand straight again after storms bend them

Answer key (try the quiz first!): 1: B 2: C 3: B 4: A 5: A 6: D

Questions to think about

1. Researchers working with INRAE: In 2012 they showed that plants have proprioception too, not only animals.

MY THOUGHTS (at least 20 words):